

1 Highlights

2 **Interpolation versus generalization in machine learning prediction** 3 **of micropollutant adsorption on layered double hydroxides**

4 [AUTHORS TO BE ADDED]

- 5 • 1,342 LDH adsorption records from 59 studies evaluated leakage-aware
- 6 • Random splits overstate ML skill: R^2 0.85 within studies, near zero
7 across
- 8 • Leave-one-study-out audit quantifies transfer for every source study
- 9 • Conformal intervals certify interpolation; new-study intervals widen 3–
10 $4\times$
- 11 • $\text{pH}-\text{pH}_{\text{pzc}}$ feature links model behavior to electrostatics

12 Interpolation versus generalization in machine learning
13 prediction of micropollutant adsorption on layered
14 double hydroxides

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16 **Abstract**

17 Micropollutants such as pharmaceuticals, pesticides, and personal-care prod-
18 ucts are detected in water bodies worldwide, and layered double hydroxides
19 (LDHs) are widely studied as tunable adsorbents for their removal. Machine
20 learning (ML) models trained on literature-compiled datasets increasingly
21 report near-perfect prediction of adsorption performance, yet the evaluation
22 protocols behind these reports rarely account for the structure of the under-
23 lying data, in which many experimental records originate from few source
24 studies. We compiled 1,342 adsorption experiments for LDH–micropollutant
25 systems from 59 published studies and evaluated seven regression models (lin-
26 ear regression, k -nearest neighbors, neural network, Cubist, random forest,
27 XGBoost, and histogram gradient boosting) under a leakage-aware protocol
28 that separates two questions: interpolation within studied systems and gen-
29 eralization to unseen studies. With duplicate records removed and all prepro-
30 cessing fitted on training folds only, the best models predicted removal ratio
31 (RR%) for studied systems at new operating conditions with R^2 of 0.85 ± 0.03
32 and RMSE near 11 percentage points across 20 repeated splits. Under leave-
33 one-study-out evaluation, however, the median per-study error rose to 14–16

percentage points, two to three times the interpolation error, and only 20 of the 59 studies were predicted with accuracy resembling the interpolation regime, showing that random-split protocols overstate the screening ability of literature-trained models by a wide margin; the models still reduce the error of descriptor-free baselines by roughly 40 percent, so some signal does transfer. Split-conformal prediction intervals certified the interpolation task (93% empirical coverage at a 90% nominal level, with ± 19 -point intervals), but for unseen studies the intervals had to widen to ± 68 points to maintain coverage, and a common distance-to-training applicability score carried no information about realized error in either task. Chemistry-informed features, including the electrostatic driving force $\text{pH}-\text{pH}_{\text{pzc}}$, changed the errors little yet yielded feature attributions consistent with electrostatic and hydrophobic adsorption mechanisms. The results delimit what literature-trained adsorption models can currently support in practice: condition optimization for studied systems, but not yet the screening of new adsorbent-pollutant combinations.

Keywords: layered double hydroxide, micropollutant, adsorption, machine learning, data leakage, uncertainty quantification

52 Nomenclature

LDH	layered double hydroxide
RR%	removal ratio (percent)
C_i	initial concentration (mg/L)
AD	adsorbent dose (g/L)
pH _{pzc}	pH at the point of zero charge
BET	Brunauer–Emmett–Teller surface area (m ² /g)
PV	pore volume (cm ³ /g)
APD	average pore diameter (nm)
d_{003}	basal interlayer spacing (nm)
53 log K_{ow}	octanol–water partition coefficient
ML	machine learning
LR, KNN, ANN	linear regression, k -nearest neighbors, neural network
RF, XGB, HistGB	random forest, XGBoost, histogram gradient boosting
LOSO	leave-one-study-out
(T1), (T2)	interpolation task, cross-study generalization task
MAE, RMSE	mean absolute error, root mean square error
SHAP	Shapley additive explanations
CV	cross-validation

54 1. Introduction

55 Micropollutants, including pharmaceuticals, pesticides, endocrine disrup-
56 tors, and personal-care products, are detected at ng/L to µg/L levels in
57 surface water, groundwater, and treated effluent worldwide (TODO, 0000d).
58 Conventional treatment trains remove many of these compounds only par-

59 tially, and their chronic ecological and human-health effects have placed
60 micropollutant control on regulatory agendas in the European Union, the
61 United States, and Asia (TODO, 0000e). Adsorption is among the most
62 widely deployed polishing steps because it requires no reagent input, pro-
63 duces no transformation by-products, and operates across wide concentration
64 ranges (TODO, 0000b).

65 Layered double hydroxides (LDHs) are a class of anionic clays with the
66 general formula $[M_{1-x}^{2+}M_x^{3+}(\text{OH})_2]^{x+}[A^{n-}]_{x/n} \cdot m\text{H}_2\text{O}$, whose metal composi-
67 tion, interlayer anion, and calcination state can be tuned to target specific
68 pollutants (TODO, 0000c). Hundreds of LDH variants have been synthesized
69 and tested against individual micropollutants, and each such study reports
70 removal under its own grid of pH, temperature, contact time, adsorbent dose,
71 and initial concentration. The resulting literature is large but fragmented: no
72 single laboratory can cover the combinatorial space of metal pairs, synthesis
73 routes, and pollutants experimentally.

74 Isotherm and kinetic models describe each of these systems in isolation
75 and offer no mechanism for pooling evidence across studies. Machine learning
76 (ML) offers a natural response to this fragmentation. By compiling published
77 experiments into a single dataset, ML models have been trained to predict
78 adsorption capacity or removal ratio from material, pollutant, and operat-
79 ing descriptors, with reported test R^2 values frequently above 0.9 (TODO,
80 0000f,g,a). Such models promise two distinct services: interpolating the be-
81 havior of an already-studied adsorbent–pollutant system at untested condi-
82 tions, and screening new systems before synthesis. The second service is the
83 one most frequently invoked to motivate these studies.

84 However, we argue that the evaluation protocols behind these reports do
85 not support the screening claim. Firstly, literature-compiled datasets are
86 strongly clustered: in our compilation, 1,342 experimental records originate
87 from only 59 source studies, and within one study the material and pollutant
88 descriptors are constant while only operating conditions vary. A random
89 record-level train–test split therefore places near-duplicate records on both
90 sides, and the test score measures within-study interpolation while being
91 reported as generalization. Secondly, preprocessing steps such as imputa-
92 tion, scaling, and feature selection are commonly fitted on the full dataset
93 before splitting, which leaks test-set information into training (Kapoor and
94 Narayanan, 2023). Thirdly, point predictions are reported without uncer-
95 tainty estimates or applicability bounds, leaving practitioners no way to know
96 when a prediction should be trusted.

97 To address the above challenges, we evaluate ML models for micropollu-
98 tant adsorption on LDHs under a protocol that closes each of these leakage
99 pathways. In essence, our idea is to treat the source study as the unit of
100 generalization: we evaluate every model twice, once under a duplicate-free
101 record-level split that answers the interpolation question, and once under
102 grouped, leave-one-study-out protocols that answer the screening question,
103 with all preprocessing confined to training folds in both cases. We comple-
104 ment point predictions with split-conformal intervals and a distance-based
105 applicability-domain score, and we examine whether chemistry-informed fea-
106 tures improve cross-study transfer.

107 More specifically, this study aims to (1) quantify the gap between inter-
108 polation and cross-study generalization for seven regression models on 1,342

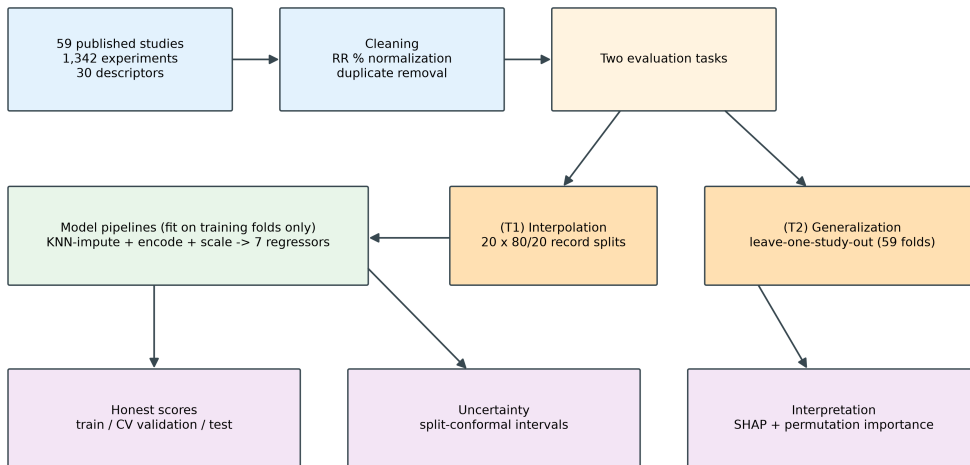


Figure 1: Evaluation workflow

LDH-micropollutant records from 59 studies; (2) equip the predictions with distribution-free uncertainty intervals and an applicability-domain flag, and test their reliability under both tasks; and (3) evaluate chemistry-informed features, including the electrostatic driving force $\text{pH}-\text{pH}_{\text{pzc}}$, and read the resulting feature attributions against known adsorption mechanisms. To the best of our knowledge, this is the first study of LDH adsorption modeling that separates these two prediction tasks explicitly and reports what literature-trained models can, and cannot yet, deliver for practice.

2. Materials and methods

Figure 1 summarizes the workflow: compilation and cleaning of the literature data, the two-task evaluation protocol, leakage-controlled model pipelines, and the uncertainty and interpretation layers.

121 2.1. Data curation

122 We compiled 1,342 batch-adsorption experiments for LDH–micropollutant
123 systems from 59 peer-reviewed studies. Each record holds 30 descriptors in
124 four groups: micropollutant properties (compound identity, $\log K_{ow}$, molecu-
125 lar weight, the Abraham solvation parameters E , S , A , B , V , and topological
126 polar surface area), LDH composition and synthesis (divalent and trivalent
127 metal, molar ratio, synthesis route, hydrothermal temperature and time, cal-
128 cination conditions, and composition-averaged elemental descriptors), mate-
129 rial texture (interlayer spacing d_{003} , BET surface area, pore volume, average
130 pore diameter, and point of zero charge pH_{pzc}), and operating conditions
131 (pH, temperature, contact time, adsorbent dose, and initial concentration).
132 Table 1 summarizes the numeric descriptors; the prediction target is the
133 removal ratio (RR%).

134 **Cleaning.** Values stored as fractions were converted to percent by a
135 range-aware rule that leaves already-percent data untouched, and category
136 labels were normalized (e.g., merging spelling variants of *co-precipitation*).
137 Records identical in every descriptor were reduced to a single occurrence,
138 since duplicates spanning a train–test boundary would let the test be an-
139 swered by lookup; deduplication removed 83 records. Missing texture val-
140 ues were imputed by a k -nearest-neighbor imputer computed in standardized
141 space and mapped back to physical units; the imputer, like every fitted trans-
142 form in this work, was trained on training folds only. The source study of
143 each record is retained as a grouping key and is never used as a model input.
144 Details of the cleaning rules are in Text S1.

145 2.2. Evaluation protocol

146 **Two prediction tasks.** We evaluate every model under two protocols
147 that correspond to the two services expected from literature-trained models.
148 Task (T1), *within-system interpolation*, asks how well a model predicts RR%
149 for a studied adsorbent–pollutant system at new operating conditions. Task
150 (T2), *cross-study generalization*, asks how well the model predicts systems it
151 has never seen. Intuitively, (T1) allows the model to have seen the material
152 and the pollutant at other conditions, while (T2) does not.

153 **Repeated splits for (T1).** A single random split can be lucky or un-
154 lucky, so (T1) is measured over 20 seeded 80/20 record-level splits of the
155 duplicate-free data, and scores are reported as mean $\pm$ standard deviation
156 across repeats. Hyperparameters are tuned once (Section 2.3) and held fixed
157 across repeats, so the spread reflects sampling variability rather than tuning
158 noise.

159 **Leave-one-study-out audit for (T2).** Each of the 59 studies s is
160 held out once; the model f_{-s} is refit on the remaining 58 studies with fixed
161 hyperparameters and scored on the held-out records:

$$\text{MAE}_s = \frac{1}{n_s} \sum_{i \in s} |y_i - f_{-s}(\mathbf{x}_i)|. \quad (1)$$

162 Reporting the distribution of MAE_s across studies, rather than one pooled
163 score, exposes which systems transfer and which do not. A learning curve
164 over the number of training studies tests whether cross-study error is limited
165 by data volume or by data diversity.

166 **Naive baselines.** To make the error magnitudes interpretable, we score
167 three predictors that use no descriptors: the global training-mean; the per-
168 pollutant mean (the average RR% of the same compound in the training

169 data, falling back to the global mean for unseen compounds); and, for (T1)
170 only, the per-study mean. A model is useful for screening only if it beats
171 these lookups on unseen studies.

172 **Leakage control.** Imputation, scaling, and categorical encoding are
173 embedded in each model pipeline and fitted on training folds only. Hyperpa-
174 rameter tuning uses cross-validation folds that differ from the folds used for
175 reporting, and grouped tasks use group-aware folds throughout. These con-
176 trols address the leakage mechanisms documented for applied ML (Kapoor
177 and Narayanan, 2023), which we found to inflate reported skill substantially
178 when absent (Section 3.3).

179 **Deployment workflow.** The protocol doubles as a reusable recipe for
180 future compilations (Fig. 1). Step 1: curate and normalize the literature
181 records, retaining the source-study identifier. Step 2: remove exact dupli-
182 cates and declare the prediction task: interpolation within studied systems or
183 screening of new ones. Step 3: split accordingly (record-level for the former,
184 study-grouped for the latter) and fit pipelines whose preprocessing sees train-
185 ing folds only. Step 4: wrap the chosen model in split-conformal calibration
186 matched to the declared task and report coverage empirically. Every step is
187 implemented in the released pipeline (`ldh_pipeline.py`), so re-running the
188 audit on an extended compilation requires only the new spreadsheet.

189 2.3. Regression models

190 We compare seven regressors spanning the model families used in the
191 adsorption-ML literature, chosen so that their inductive biases differ. Through-
192 out, $\mathbf{x}_i \in \mathbb{R}^p$ denotes the preprocessed descriptor vector of record i and $\hat{y}_i$ its
193 predicted RR%.

194 **Linear regression (LR)** fits one additive coefficient per descriptor,

$$\hat{y} = \beta_0 + \sum_{j=1}^p \beta_j x_j, \quad (2)$$

195 with β estimated by least squares. It serves as the interpretable floor: any
 196 nonlinearity or interaction in the adsorption response appears as a gap be-
 197 tween LR and the flexible models.

198 **k -nearest neighbors (KNN)** predicts the average target of the k closest
 199 training records in standardized feature space (Cover and Hart, 1967),

$$\hat{y}(\mathbf{x}) = \frac{1}{k} \sum_{i \in \mathcal{N}_k(\mathbf{x})} y_i, \quad (3)$$

200 where $\mathcal{N}_k(\mathbf{x})$ holds the k nearest neighbors under Euclidean distance and
 201 k is grid-searched between 1 and 9. KNN has no functional form at all;
 202 its performance measures how far pure similarity lookup carries prediction,
 203 which makes it a diagnostic for near-duplicate structure in the data.

204 **Artificial neural network (ANN)** is a single-hidden-layer perceptron
 205 trained by Adam (Rumelhart et al., 1986),

$$\hat{y} = \mathbf{w}^\top \sigma(W\mathbf{x} + \mathbf{b}) + b_0, \quad (4)$$

206 with 32 hidden units and rectified-linear activation $\sigma(z) = \max(0, z)$, match-
 207 ing the compact architectures used in prior adsorption studies. Larger archi-
 208 tectures were not explored because the dataset holds fewer than 10^3 effective
 209 degrees of freedom per feature.

210 **Cubist** builds a set of rules R_m whose leaves hold linear models (Quinlan,
 211 1992),

$$\hat{y}(\mathbf{x}) = \text{avg}_{m: \mathbf{x} \in R_m} \left(\beta_0^{(m)} + \sum_j \beta_j^{(m)} x_j \right), \quad (5)$$

optionally smoothed toward the targets of nearby training instances; committees and neighbors are grid-searched. Unlike constant-leaf trees, its leaf-linear structure can extrapolate moderately beyond the training range of a rule, a property that becomes relevant under (T2).

Random forest (RF) averages B decorrelated trees grown on bootstrap samples (Breiman, 2001),

$$\hat{g}(\mathbf{x}) = \frac{1}{B} \sum_{b=1}^B T_b(\mathbf{x}), \quad (6)$$

providing a variance-reduced piecewise-constant baseline among the ensembles.

Gradient-boosted trees build the prediction stagewise, each new tree fitting the residual structure left by its predecessors,

$$F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \nu h_m(\mathbf{x}), \quad (7)$$

with learning rate ν and base trees h_m . **XGBoost** implements Eq. (7) with second-order updates and a regularized objective that penalizes the number of leaves and their weights (Chen and Guestrin, 2016); **histogram gradient boosting (HistGB)** is the scikit-learn implementation with histogram-binned split finding (Ke et al., 2017; Pedregosa et al., 2011). Both are the strongest current defaults for tabular data.

Tree-ensemble hyperparameters were tuned with the TPE sampler of Optuna over 30 trials (Akiba et al., 2019) on cross-validation folds of the training data. Scale-sensitive models (LR, KNN, ANN, Cubist) receive a standardizer inside their pipeline; tree ensembles operate on raw feature scales. Search spaces and selected values are in Table S1; all splits, folds, and samplers are seeded for reproducibility.

234 2.4. Feature screening and selection

235 Multicollinearity is diagnosed with the variance inflation factor,

$$\text{VIF}_j = \frac{1}{1 - R_j^2}, \quad (8)$$

236 where R_j^2 is the coefficient of determination of descriptor j regressed on all
 237 other descriptors (targets excluded). Because descriptors are constant within
 238 a study, the study-cluster structure drives VIF values orders of magnitude
 239 beyond the usual threshold of 10 (Section 3.1), so VIF is used here as a
 240 redundancy diagnostic rather than an automatic elimination rule.

241 Feature selection is then evaluated the same way as the models them-
 242 selves: three nested descriptor sets (the full set of 30, a correlation-screened
 243 set of 21 that drops the collinear Abraham and composition-averaged block,
 244 and the 10 descriptors with the best median permutation-importance rank)
 245 are compared under both tasks with fixed hyperparameters. Judging a fea-
 246 ture set only under (T1) would inherit the same optimism as judging a model
 247 there, so both tasks are reported.

248 2.5. Uncertainty and applicability domain

249 **Split-conformal intervals.** We wrap the best model in split-conformal
 250 prediction (Vovk et al., 2005; Lei et al., 2018). A calibration subset of n
 251 records is withheld from training, absolute residuals $r_i = |y_i - \hat{y}_i|$ are com-
 252 puted on it, and the interval half-width is their conformal quantile

$$q = \text{Quantile}\left(\{r_i\}_{i=1}^n; \frac{\lceil (n+1)(1-\alpha) \rceil}{n}\right), \quad (9)$$

253 so that each test prediction is reported as $[\hat{y} - q, \hat{y} + q]$ with nominal coverage
 254 $1 - \alpha = 0.9$. For task (T2) the calibration subset is drawn as whole studies,

so the calibration residuals reflect the new-study setting. Conformal validity assumes exchangeability between calibration and test units, which grouped data satisfy only approximately; we therefore report empirical coverage for both tasks rather than assuming it.

Applicability domain. As a candidate trust flag we compute, for each test record, the mean standardized distance to its five nearest training records in the preprocessed feature space. Records far from the training distribution should receive predictions with caution; we validate the flag by correlating it with realized absolute error rather than assuming it works.

2.6. Feature engineering and attribution

Beyond the raw descriptors, we construct three features with mechanistic meaning: the electrostatic driving force $\text{pH}-\text{pH}_{\text{pzc}}$, whose sign determines whether the LDH surface is positively or negatively charged relative to the solution; the initial loading ratio C_i/dose ; and the interaction $\log K_{\text{ow}} \times (\text{pH}-\text{pH}_{\text{pzc}})$ coupling hydrophobicity with surface charge. Both tasks are re-evaluated with and without these features. Feature attributions are computed with SHAP (Lundberg and Lee, 2017) on the held-out test set using the fitted models, complemented by permutation importance ranks across all seven models and one-dimensional partial dependence profiles, and the resulting patterns are read against the expected electrostatic and hydrophobic mechanisms.

2.7. Performance metrics

Model skill is reported as the coefficient of determination,

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}, \quad (10)$$

the root mean square error, $\text{RMSE} = \sqrt{\frac{1}{n} \sum_i (y_i - \hat{y}_i)^2}$, and the mean absolute error, $\text{MAE} = \frac{1}{n} \sum_i |y_i - \hat{y}_i|$, each computed on three bases with distinct meanings: the training records (in-sample fit), cross-validation folds of the training set (validation), and the held-out test records of the task at hand. All three are reported to keep the in-sample ceiling and the generalization estimate visibly separate. Wall-clock training and prediction times are reported in Section 3.8.

3. Results and discussion

Section 3.1 first characterizes the compiled dataset. The modeling experiments then answer four research questions. (RQ1) How accurately do the models predict RR% for studied systems at new operating conditions? (RQ2) Does this accuracy transfer to systems from unseen studies? (RQ3) Can the uncertainty of individual predictions be quantified reliably under both tasks? (RQ4) Which factors govern the predictions, and are they chemically plausible?

3.1. Dataset characteristics

Figure 2 summarizes the compiled data. The target is broad and near-symmetric (RR% mean 45.5, median 43.7, SD 28.0, skewness 0.11) and spans the full 0–100% range, but its distribution differs strongly between pollutants and between LDH metal pairs (Fig. 2a,b), reflecting both genuine chemistry and the uneven attention different systems have received in the literature. The compilation covers 14 micropollutants, 13 metal pairs, and 4 synthesis routes, with record and study counts diverging sharply: diclofenac contributes 304 records from only 3 studies, while ciprofloxacin contributes 181

records spread over 32 studies. The study-size distribution is likewise skewed: the largest single study contributes 220 of the 1,342 records, the median study only 17, and a random record-level sample is therefore dominated by a handful of intensively measured systems. Several operating variables have heavy right tails (skewness 5–6 for dose, initial concentration, and contact time; Table S4), which the rank-based correlation analysis below accommodates.

Univariate association with the target. Table 2 lists the Spearman correlation of each numeric descriptor with RR%. No descriptor exceeds $|\rho| = 0.43$ (pore volume), and the leading operating variables reach only 0.27–0.33, so no single factor controls removal monotonically. This is because the response is governed by interactions (the same pH raises removal for one pollutant–material pair and lowers it for another), which is what motivates flexible learners over univariate screening in the first place, and it cautions against reading the univariate signs mechanistically.

Subgroup contrasts. Bootstrap confidence intervals on median RR% differences (5,000 resamples; Table S9) separate real subgroup effects from compilation artifacts. Calcined materials remove a median 10.1 points more than uncalcined ones (95% CI [1.7, 17.9]), and diclofenac records exceed tetracycline records by 15.5 points (CI [8.3, 27.3]); both intervals exclude zero. The apparent Zn–Al advantage over Mg–Al (8.7 points), by contrast, is not significant (CI [−0.7, 17.2]), so metal-pair claims from this compilation would be premature.

Spearman correlations among the numeric descriptors (Fig. S1) show the expected blocks: the Abraham solvation parameters correlate strongly with molecular weight and with each other, and the composition-averaged ele-

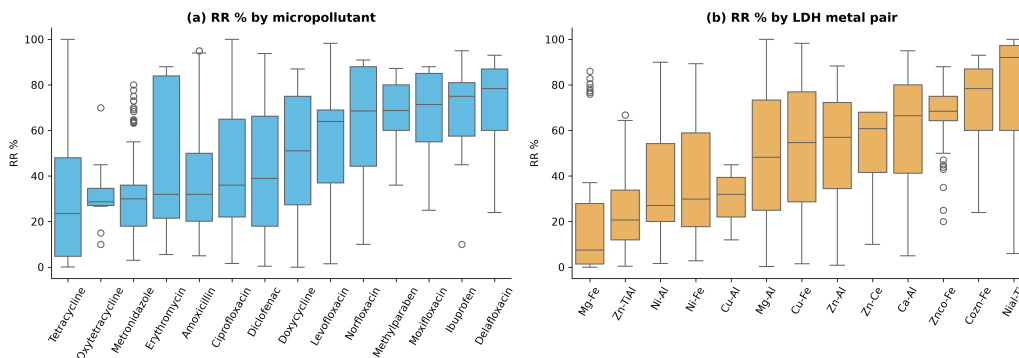


Figure 2: Removal-ratio distributions by pollutant and by metal pair

mental descriptors correlate with the metal identities they are derived from. Variance inflation factors make the redundancy explicit: 22 of 26 numeric descriptors exceed the usual threshold of 10, with a median VIF near 7×10^4 (Table S6), because descriptors are constant within a study, so the study-cluster structure induces near-perfect collinearity. This redundancy motivates the feature-selection experiment of Section 3.4 and, more fundamentally, explains why record-level splits leak: any two records of the same study are near-duplicates in feature space. Missingness is confined to the synthesis and texture blocks (Table 1): calcination conditions are unreported for two thirds of the records, and pH_{pzc} and pore volume for roughly a quarter, which is why imputation quality matters and why texture-based mechanistic claims from compiled data warrant caution.

3.2. Within-system interpolation (RQ1)

Table 3 answers (RQ1) by comparing the seven models and the naive baselines under task (T1), reported as mean $\pm$ SD over 20 seeded splits. The tree ensembles lead: XGBoost and HistGB reach test R^2 of 0.84–0.85

343 with RMSE near 11 percentage points, followed by Cubist and RF, while LR
 344 trails far behind. This is because RR% responds nonlinearly to the operating
 345 conditions, with thresholds in pH and dose that additive linear terms cannot
 346 represent. The baselines put these numbers in perspective. The per-study
 347 mean, which uses no descriptors at all, already attains R^2 of 0.43 (MAE
 348 16.7): nearly half of the variance a record-level split appears to explain is
 349 available from system identity alone, and LR barely improves on it. The
 350 ensembles' further gain to R^2 of 0.85 therefore reflects genuine modeling of
 351 the operating conditions within known systems. The in-sample R^2 of the
 352 ensembles approaches 0.99, and the gap to the test score is the expected
 353 overfitting margin of high-capacity learners rather than a protocol artifact;
 354 the cross-validation and test scores agree closely. The two leading ensembles
 355 are statistically indistinguishable: a paired Wilcoxon test on MAE across
 356 the 20 splits does not separate HistGB from XGBoost ($p = 0.23$), so we
 357 treat them as co-leaders. Adjusting R^2 for the 30 predictors changes little
 358 (0.83 ± 0.03 for HistGB versus the unadjusted 0.85). Figure 3 adds the bias
 359 diagnostics on the reference split: the fitted test regression is $\hat{y} = 0.87y + 5.6$,
 360 the mild slope shrinkage typical of regularized learners (extreme removals are
 361 pulled toward the mean), while the training fit is essentially unbiased (slope
 362 0.98). It is worth noting that these scores are already 2–3 RMSE points
 363 more conservative than a protocol without duplicate removal on the same
 364 data, since deduplication alone eliminates records whose test values could be
 365 looked up verbatim from training.

366 *3.3. Cross-study generalization (RQ2)*

367 Figure 4 answers (RQ2) with the leave-one-study-out audit. The picture
368 changes sharply: the median per-study MAE of the best models rises to 14–
369 16 percentage points, two to three times their interpolation error, and the
370 per-study R^2 is positive for only 4 of the 32 studies large enough to evaluate
371 it. This is because the models exploit descriptor combinations that identify
372 a material–pollutant system rather than transferable structure–activity rela-
373 tions; when all records of a system are withheld, that shortcut disappears.
374 The baseline comparison adds an important nuance. Descriptor-free lookups
375 transfer even worse: the global training-mean reaches a median per-study
376 MAE of 24 points and the per-pollutant mean of 25, so the models do reduce
377 the naive error by roughly 40 percent, and Cubist beats the per-pollutant
378 lookup in 41 of the 59 studies. The models therefore track the level differences
379 between systems while failing to resolve the condition response within an un-
380 seen system, which is exactly why the per-study R^2 (measured against each
381 study’s own variance) stays negative even as the MAE gain over the baselines
382 is large. Screening applications get a coarse ranking signal from these mod-
383 els, not engineering precision. The per-study distribution in Fig. 4b is wide
384 rather than uniformly poor: for Cubist, 20 of the 59 studies are predicted
385 with errors comparable to the interpolation regime (MAE below 6 points),
386 while 9 studies exceed 30 points. Notably, the model ranking also reorders:
387 Cubist, mid-table under interpolation, attains the lowest cross-study median,
388 which could be explained by its rule-plus-linear structure extrapolating more
389 gracefully than deep tree ensembles.

390 **Which systems transfer.** Figure 6 breaks the per-study errors down

by chemistry, and the pattern follows literature density. Ciprofloxacin studies, which dominate the compilation (32 of 59 studies), are predicted with a median MAE of 5 points, and Zn–Al systems (28 studies) with 5 points, because each held-out study of these families leaves many close relatives in training. Sparsely covered chemistries fare far worse: tetracycline, diclofenac, and levofloxacin studies show median errors above 20 points, and Ni–Fe systems above 30. Error also grows with the held-out study’s own scope: per-study MAE correlates with both the number of records and the within-study spread of RR% ($r \approx 0.45$ for each), meaning studies that explore wide condition ranges are precisely the ones an outside model cannot resolve. A pollutant-by-model matrix of the same audit (Table S11) shows in addition that no model wins uniformly: XGBoost leads for ciprofloxacin, Cubist for doxycycline and tetracycline, and KNN, pure similarity lookup, for amoxicillin, underlining how class-dependent cross-study prediction currently is. The learning curve in Fig. 5b is flat between 5 and 40 training studies, suggesting that cross-study error is limited by the diversity of compiled systems rather than by record count. This supports our central argument that random-split scores, including those in Table 3, must not be read as screening skill: the same models, the same data, and the same preprocessing yield near-zero skill when the test set contains genuinely new systems.

3.4. Feature selection under both tasks

Table 4 evaluates the three nested descriptor sets of Section 2.4 under both tasks with fixed hyperparameters. Selection is close to neutral in both directions. Interpolation accuracy is indistinguishable across the sets (R^2 of 0.85 ± 0.03 for HistGB in all three; XGBoost and Cubist behave the same,

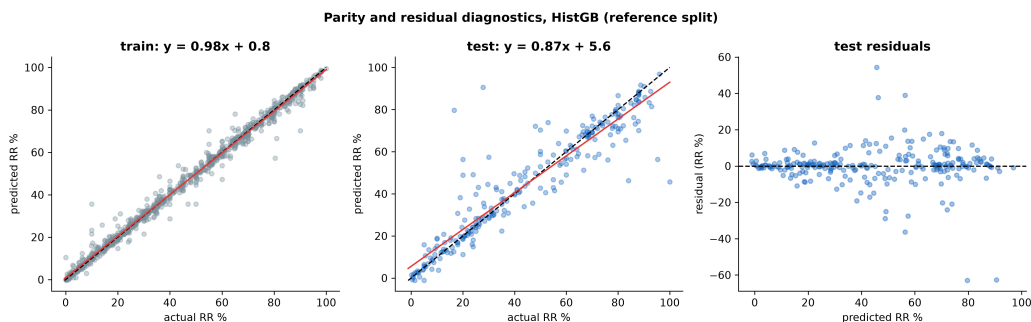


Figure 3: Parity and residual diagnostics of the best model under interpolation, with fitted slope and intercept

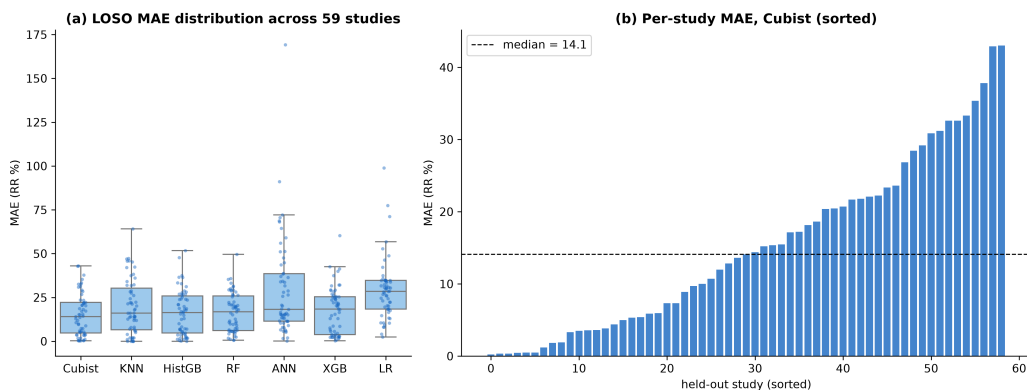


Figure 4: Leave-one-study-out audit across 59 studies

416 Table S8), so ten well-chosen descriptors carry essentially all the within-
 417 system signal of the full thirty. This is because the removable descriptors are
 418 redundant with the retained ones (the collinear Abraham and composition-
 419 averaged block flagged by the VIF analysis) rather than independently infor-
 420 mative. In the other direction, no reduction rescues cross-study transfer: the
 421 LOSO median moves only from 16.4 to 16.2 points, consistent with the flat
 422 learning curve: the transfer bottleneck is study diversity, not dimensionality.
 423 The practical reading favors parsimony: the top-10 set consists of the five

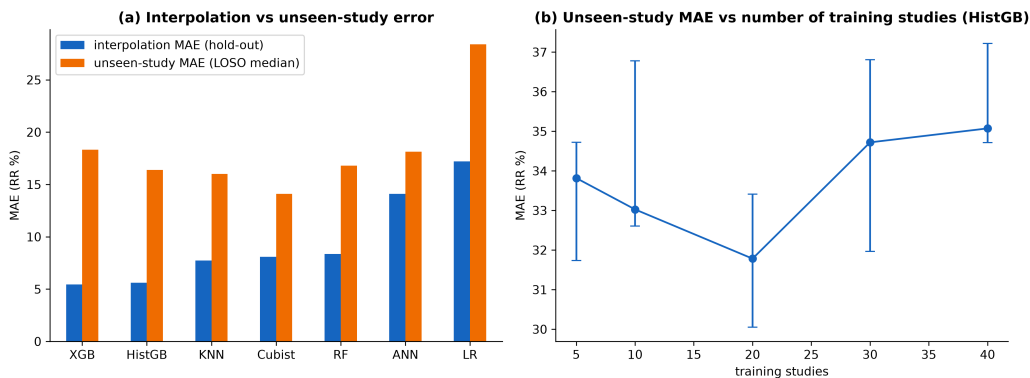


Figure 5: Interpolation versus unseen-study error and learning curve

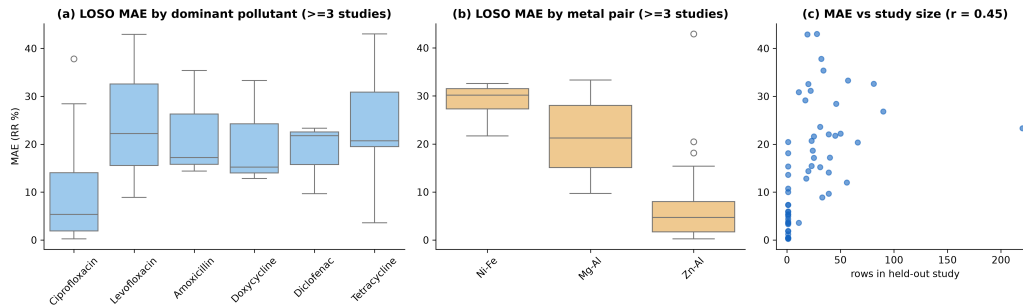


Figure 6: Cross-study error by pollutant, metal pair, and study size

operating conditions, four texture properties, and $\log K_{ow}$, all routinely reported quantities, so a deployable model does not require the full descriptor battery.

3.5. Uncertainty and applicability domain (RQ3)

Figure 7 answers (RQ3). Under interpolation, the split-conformal 90% intervals of Eq. (9) achieve empirical coverage of 93% with a half-width of 19 percentage points, so predictions for studied systems can be certified at a practically useful precision. Under the new-study task, coverage is maintained (99%) only because the study-grouped calibration inflates the half-

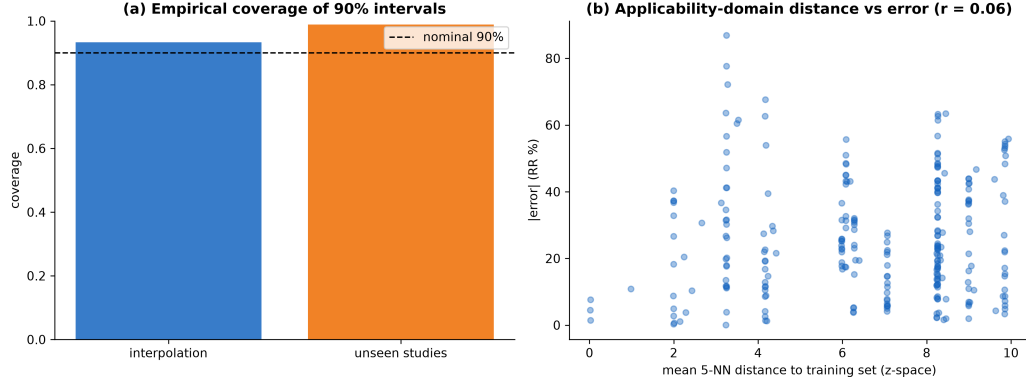


Figure 7: Conformal coverage and applicability-domain flag

width to 68 points, an interval spanning most of the 0–100 RR% range and therefore carrying little decision value. This is because residual magnitudes differ systematically between systems, so a single global quantile must cover the worst of them. The distance-based applicability-domain flag does not compensate: the mean distance to the five nearest training records carries essentially no information about realized error in either task ($r \leq 0.06$, Fig. 7b), although such distance heuristics are widely used as trust flags. A key finding is therefore that the uncertainty tools inherit the transfer problem itself: interpolation predictions can be certified, whereas no construction we tested produces informative guarantees for unseen systems, and coverage claims must be validated empirically for the deployment task.

3.6. Error tolerance and screening use

Figure 8 translates the error metrics into the terms a user experiences: the share of test predictions falling within fixed tolerance bands. Under interpolation, 84% of HistGB predictions land within 10 RR% points of the measured value and 95% within 20 points, so a practitioner optimizing con-

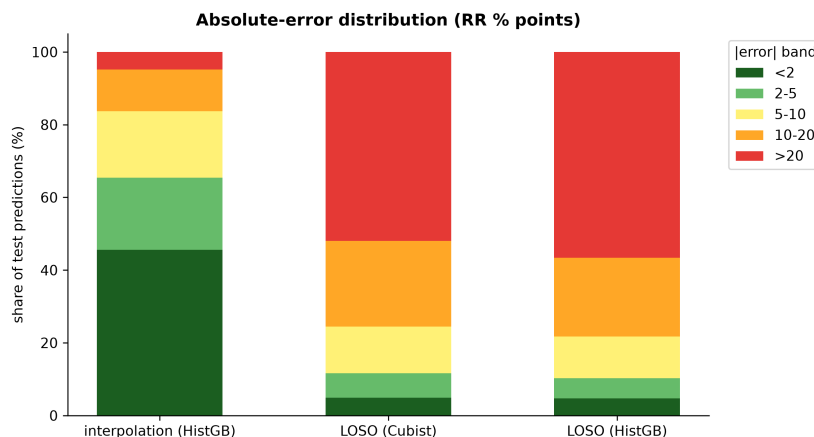


Figure 8: Distribution of absolute prediction errors by tolerance band

449 ditions for a studied system can expect single-digit errors most of the time.
 450 Under leave-one-study-out prediction the distribution inverts: only about a
 451 quarter of predictions fall within 10 points and more than half miss by over
 452 20, for Cubist and HistGB alike.

453 The screening recast makes the consequence concrete. Binarizing at $RR\%$
 454 ≥ 80 ("effective removal") and asking whether the model's shortlist is right,
 455 interpolation predictions achieve a precision of 0.92 at a recall of 0.58; the
 456 shortlist is trustworthy though incomplete. For unseen studies the same
 457 construction collapses to a precision of 0.32 and a recall of 0.06 (Table S10):
 458 of 203 genuinely effective records, the cross-study model retrieves twelve. A
 459 model whose regression does not transfer does not become useful by lowering
 460 the question to a binary shortlist; this closes the argument of Sections 3.3–3.5
 461 from the decision side.

462 3.7. *Feature effects and mechanistic reading (RQ4)*

463 Figure 9 and Table 5 answer (RQ4). Adding the chemistry-informed fea-
464 tures leaves both tasks essentially unchanged (interpolation MAE 5.6 versus
465 5.8 points; cross-study median 16.4 versus 16.0), indicating that the models
466 already reconstruct these combinations internally from pH, pH_{pzc} , and dose.
467 Their value here is interpretive rather than predictive.

468 **Operating conditions dominate.** Permutation-importance ranks on
469 the hold-out set (Table 5) agree across six of the seven models on the leading
470 factors: adsorbent dose ranks first, followed by contact time, initial concen-
471 tration, and pH, with the first material property (BET) at median rank six
472 and the first pollutant descriptor ($\log K_{\text{ow}}$) at rank nine; only the additive LR
473 deviates. The SHAP summary (Fig. 9, top) shows the same hierarchy from
474 a second attribution method, and aggregating mean $|\text{SHAP}|$ by descriptor
475 group makes it quantitative: operating conditions account for 54% of the to-
476 tal attribution mass, material texture for 18%, composition and synthesis for
477 14%, and pollutant descriptors for 14% (Table S7). The partial-dependence
478 profiles of the three leading features (Fig. S2) show textbook adsorption be-
479 havior: removal rises steeply with dose up to roughly 2–5 g/L and then
480 saturates, rises within the first hours of contact and plateaus at equilibrium,
481 and declines with initial concentration beyond roughly 200 mg/L as fixed
482 capacity is exhausted. Within known systems, the models therefore learn
483 genuine condition–response physics. The same ranking, however, explains
484 the transfer failure of Section 3.3 from a second direction: material and pol-
485 lutant descriptors, the only information available about an unseen system,
486 contribute comparatively little to the fitted models.

487 **Mechanistic reading.** The SHAP dependence of $\text{pH}-\text{pH}_{\text{pzc}}$ (Fig. 9a)
 488 is not smoothly monotone but threshold-like: contributions stay within a
 489 few points while the surface remains positively charged, and a sharp penalty
 490 of up to nine points appears once solution pH exceeds pH_{pzc} by more than
 491 about three units. Coloring this dependence by $\log K_{\text{ow}}$ (Fig. S3) shows the
 492 penalty concentrated among hydrophilic compounds, consistent with elec-
 493 trostatic repulsion of anionic species from a negatively charged surface while
 494 hydrophobic partitioning buffers the loss for lipophilic ones. $\log K_{\text{ow}}$ itself
 495 is the strongest chemical attribution (Fig. 9b), monotone from roughly -5
 496 points for the most hydrophilic compounds to $+15$ for the most lipophilic.
 497 BET surface area (Fig. 9c) is a cautionary counterexample: the model assigns
 498 positive contributions to records with low measured BET even though the
 499 raw association runs the other way (median RR% of 44 below $20 \text{ m}^2/\text{g}$ versus
 500 50 above), so this conditional attribution reflects the correlation structure of
 501 the compilation rather than a surface-area mechanism, and we do not inter-
 502 pret it mechanistically. Attributions for the composition-averaged elemental
 503 descriptors are small, consistent with the finding that current composition
 504 descriptors carry little transferable information between studies. [MECHA-
 505 NISM PARAGRAPH FLAGGED FOR DOMAIN REVIEW BY THE AU-
 506 THORS.]

507 3.8. Computation time

508 Training cost is not a differentiator at this data scale. On the reference
 509 split, a single fit takes below one second for every model except XGBoost (7–
 510 10 s across repeated measurements with the tuned 495-tree configuration),
 511 and prediction takes below 0.1 s throughout (Table S3). The full leave-one-

study-out audit, 59 refits of all seven models, completes within minutes on a workstation, so the honest grouped protocol carries no meaningful computational premium over the random-split protocol it replaces. Hyperparameter search with Optuna dominates the overall budget and is performed once, not per evaluation task.

3.9. Comparison with previous studies

Table 6 places this work among prior ML compilations for adsorption prediction. To the best of our knowledge, none of the prior compilations reports a study-grouped evaluation, so their published scores are comparable to our record-level column rather than to the screening column. Our record-level numbers (R^2 of 0.85 ± 0.03) sit inside the range those studies report, which supports rather than contradicts them; the point of the comparison is that all such scores, ours included, measure interpolation within compiled systems, and the grouped column shows how much smaller the transferable part is. Re-evaluating published models under grouped protocols would clarify how general this pattern is across adsorbent families.

3.10. Environmental implications and outlook

For practice, the results delimit two use cases sharply. Water-treatment engineers optimizing operating conditions for an LDH-pollutant system that has been measured before can rely on literature-trained models with a mean absolute error near 6 percentage points of RR%, tightened further by conformal intervals. Materials researchers seeking to screen new LDH formulations or new pollutants face a harder trade: the models do carry transferable signal, reducing naive baseline errors by roughly 40 percent, but per-study skill

536 remains near zero, so predictions for unmeasured systems support coarse pri-
537 oritization at best and should be treated as hypotheses for experiment rather
538 than estimates.

539 **Recommendations by stakeholder.** For treatment operators: literature-
540 trained models are ready for condition optimization of measured systems,
541 and the conformal intervals of Section 3.5 supply defensible error bars for
542 that use. For modelers: report study-grouped scores alongside record-level
543 ones: the two-task protocol here adds minutes of computation (Section 3.8)
544 and prevents the largest overstatement this field currently makes. For ex-
545 perimentalists and data curators: report interlayer anion identity, surface
546 morphology, and standardized pH_{pzc} /BET protocols, since these unreported
547 properties are the most plausible carriers of the transferable signal that cur-
548 rent descriptors lack; and prioritize new *systems* over additional condition
549 points on studied ones, because the learning curve shows cross-study error is
550 limited by study diversity, not record count. The same logic governs model
551 maintenance: retraining on an extended compilation is worthwhile when new
552 studies arrive, and the released pipeline re-runs the full audit on an updated
553 spreadsheet without code changes.

554 **Choice of target variable.** $\text{RR}\%$ is a condition-bound outcome rather
555 than an intrinsic material property, which makes it honest for operations but
556 couples every prediction to dose, time, and concentration. Capacity-style
557 targets such as Q_e or the Langmuir-derived Q_m look more transferable in
558 principle, but they inherit the fitting error of the isotherm model that pro-
559 duced them, and in our original-dataset experiments Q_e prediction showed
560 the same gap in amplified form: Cubist reached a test R^2 of 0.98 under in-

561 terpolation yet a leave-one-study-out median MAE of 42 mg/g, with positive
562 per-study R^2 in only 3 of 32 evaluable studies. The evaluation protocol, not
563 the target choice, determines whether reported skill is real.

564 We see three concerns that temper the conclusions. First, the compila-
565 tion covers 59 studies, so the cross-study estimates themselves carry sampling
566 uncertainty; the per-study distribution, not a single mean, is the honest sum-
567 mary. The second concern is reporting heterogeneity between laboratories
568 (differing protocols for pH_{pzc} or BET), which acts as irreducible noise under
569 (T2) and may partly explain the transfer gap. The third concern is that our
570 descriptors, like those of prior compilations, describe the bulk material; in-
571 terlayer anion identity and surface morphology are rarely reported and may
572 carry the transferable signal that current models lack. In future work, richer
573 composition-structure descriptors, study-level random effects, and few-shot
574 calibration on a handful of measurements of a new system are the most
575 promising routes toward genuine screening capability.

576 4. Conclusions

577 We evaluated seven machine learning models for micropollutant adsorp-
578 tion on layered double hydroxides under a leakage-aware protocol that sepa-
579 rates within-system interpolation from cross-study generalization, using 1,342
580 experiments compiled from 59 studies. The models predict removal ratio for
581 studied systems at new operating conditions with test R^2 of 0.85 ± 0.03
582 and RMSE near 11 percentage points across 20 repeated splits, but the same
583 models show near-zero per-study skill when entire studies are held out, and a
584 leave-one-study-out audit locates the failure in a wide per-study error distri-

585 bution rather than a uniform degradation. Against descriptor-free baselines
586 the picture is more nuanced: the models reduce the naive cross-study error
587 by roughly 40 percent, so they do carry transferable signal, yet not enough
588 to certify predictions for an unmeasured system. Split-conformal intervals
589 certify interpolation predictions at ± 19 percentage points, whereas for un-
590 seen studies neither conformal calibration nor a distance-based applicability
591 score yields informative guarantees. Chemistry-informed features such as
592 $\text{pH} - \text{pH}_{\text{pzc}}$ leave the accuracy essentially unchanged but anchor the learned
593 behavior in electrostatic and hydrophobic mechanisms. In future work, we
594 plan to extend the compilation with interlayer-anion and morphology de-
595 scriptors, model study-level effects explicitly, and test few-shot calibration
596 so that a small number of measurements of a new system can recover the
597 accuracy currently reserved for studied ones.

598 **CRedit authorship contribution statement**

599 [TO BE COMPLETED BY THE AUTHORS.]

600 **Declaration of competing interest**

601 The authors declare that they have no known competing financial inter-
602 ests or personal relationships that could have appeared to influence the work
603 reported in this paper.

604 **Data availability**

605 The curated dataset, the full modeling pipeline (`ldh_pipeline.py`), and
606 the notebooks reproducing every table and figure are available at [REPOSI-

607 TORY URL TO BE ADDED].

608 **References**

- 609 Akiba, T., Sano, S., Yanase, T., Ohta, T., Koyama, M., 2019. Optuna: A
610 next-generation hyperparameter optimization framework, in: Proceedings
611 of the 25th ACM SIGKDD International Conference on Knowledge Dis-
612 covery and Data Mining, pp. 2623–2631. doi:10.1145/3292500.3330701.
- 613 Breiman, L., 2001. Random forests. Machine Learning 45, 5–32. doi:10.
614 1023/A:1010933404324.
- 615 Chen, T., Guestrin, C., 2016. XGBoost: A scalable tree boosting sys-
616 tem, in: Proceedings of the 22nd ACM SIGKDD International Conference
617 on Knowledge Discovery and Data Mining, pp. 785–794. doi:10.1145/
618 2939672.2939785.
- 619 Cover, T., Hart, P., 1967. Nearest neighbor pattern classification. IEEE
620 Transactions on Information Theory 13, 21–27. doi:10.1109/TIT.1967.
621 1053964.
- 622 Kapoor, S., Narayanan, A., 2023. Leakage and the reproducibility crisis
623 in machine-learning-based science. Patterns 4, 100804. doi:10.1016/j.
624 patter.2023.100804.
- 625 Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., Liu,
626 T.Y., 2017. LightGBM: A highly efficient gradient boosting decision tree,
627 in: Advances in Neural Information Processing Systems 30, pp. 3146–3154.

628 Lei, J., G'Sell, M., Rinaldo, A., Tibshirani, R.J., Wasserman, L., 2018.
 629 Distribution-free predictive inference for regression. *Journal of the Ameri-*
 630 *can Statistical Association* 113, 1094–1111. doi:10.1080/01621459.2017.
 631 1307116.

632 Lundberg, S.M., Lee, S.I., 2017. A unified approach to interpreting model
 633 predictions, in: *Advances in Neural Information Processing Systems* 30,
 634 pp. 4765–4774.

635 Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel,
 636 O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J.,
 637 Passos, A., Cournapeau, D., Brucher, M., Perrot, M., Duchesnay, É., 2011.
 638 Scikit-learn: Machine learning in Python. *Journal of Machine Learning*
 639 *Research* 12, 2825–2830.

640 Quinlan, J.R., 1992. Learning with continuous classes, in: *Proceedings of the*
 641 *5th Australian Joint Conference on Artificial Intelligence*, pp. 343–348.

642 Rumelhart, D.E., Hinton, G.E., Williams, R.J., 1986. Learning represen-
 643 tations by back-propagating errors. *Nature* 323, 533–536. doi:10.1038/
 644 323533a0.

645 TODO, 0000a. [todo: Acs est water biochar nutrient adsorption ml paper
 646 (in user's collection: acsestwater.2c00301) — extract real citation from the
 647 pdf].

648 TODO, 0000b. [todo: adsorption for micropollutant removal — review].

649 TODO, 0000c. [todo: Ldh synthesis/applications review].

650 TODO, 0000d. [todo: micropollutant occurrence in water bodies — global
651 review].

652 TODO, 0000e. [todo: micropollutant regulation eu/us/asia].

653 TODO, 0000f. [todo: Ml for adsorption prediction — example 1].

654 TODO, 0000g. [todo: Ml for adsorption prediction — example 2].

655 Vovk, V., Gammerman, A., Shafer, G., 2005. Algorithmic Learning in a
656 Random World. Springer. doi:10.1007/b106715.

Table 1: Summary statistics of the numeric descriptors

Descriptor	Median	Range	Missing (%)
<i>Micropollutant</i>			
$\log K_{ow}$	−0.02	−2.67–4.51	0
MW (g/mol)	331	152–734	0
$E / S / A / B / V$	1.81 / 2.10 / 0.70 / 1.70 / 2.30	–	0
TPSA (Å ²)	76.2	37.3–194.0	0
<i>Composition and synthesis</i>			
M(II)/M(III) ratio	3.0	0.5–8.0	0
Avg. atomic weight (g/mol)	48.1	24.8–73.7	0
Avg. d-electrons	4.25	0–8.89	0
Avg. Pauling electronegativity	1.59	1.30–1.89	0
Hydrothermal temp. (°C)	70	25–160	13.3
Hydrothermal time (h)	24	12–40	13.3
Calcination temp. (°C)	400	200–550	65.9
Calcination time (h)	4	1–12	65.9
<i>Material texture</i>			
d_{003} (nm)	0.77	0.25–1.13	2.5
BET (m ² /g)	82.0	0.4–225.7	7.9
PV (cm ³ /g)	0.10	0–1.46	25.3
APD (nm)	8.4	0.2–64.2	13.9
pH _{pzc}	7.5	5.5–11.6	24.6
<i>Operating conditions</i>			
pH	6.7	2.0–12.0	0
Temperature (°C)	25	10–55	0
Contact time (h)	1.5	0.02–192	0
Dose (g/L)	0.9	0.01–50	0
C_i (mg/L)	40	1–1688	0

Table 2: Spearman rank correlation of numeric descriptors with RR% (twelve largest by magnitude), with bootstrap 95% confidence intervals (2,000 resamples); every interval excludes zero

Descriptor	ρ	95% CI
PV (cm ³ /g)	0.43	[0.37, 0.47]
Dose (g/L)	0.33	[0.28, 0.38]
E	-0.30	[-0.35, -0.25]
APD (nm)	0.28	[0.22, 0.34]
pH	0.27	[0.21, 0.32]
Hydrothermal time (h)	0.23	[0.18, 0.29]
TPSA	-0.19	[-0.25, -0.14]
S	-0.19	[-0.24, -0.13]
C_i (mg/L)	0.18	[0.13, 0.23]
d_{003} (nm)	0.17	[0.11, 0.22]
B	-0.16	[-0.22, -0.10]
Calcination time (h)	0.16	[0.06, 0.27]

Table 3: Model and baseline performance under within-system interpolation (T1), mean $\pm$ SD over 20 seeded 80/20 splits

Model	Test R^2	Test MAE	Test RMSE
XGBoost	0.84 ± 0.03	6.3 ± 0.6	11.1 ± 1.0
HistGB	0.85 ± 0.03	6.3 ± 0.5	10.8 ± 1.0
KNN	0.73 ± 0.06	7.8 ± 0.8	14.4 ± 1.5
Cubist	0.79 ± 0.03	8.0 ± 0.6	12.7 ± 1.0
RF	0.78 ± 0.03	8.6 ± 0.5	13.0 ± 0.9
ANN	0.54 ± 0.05	14.4 ± 0.7	19.1 ± 0.9
LR	0.40 ± 0.04	17.4 ± 0.5	21.7 ± 0.7
Per-study mean	0.43 ± 0.05	16.7 ± 0.8	21.2 ± 0.9
Per-pollutant mean	0.18 ± 0.03	21.4 ± 0.6	25.4 ± 0.6
Global mean	-0.01 ± 0.01	24.9 ± 0.7	28.1 ± 0.6

Table 4: Feature-set comparison under both tasks (HistGB; interpolation mean $\pm$ SD over 20 splits, LOSO median over 59 studies)

Feature set	p	Interp. R^2	Interp. MAE	LOSO median MAE
Full	30	0.85 ± 0.03	6.3 ± 0.5	16.4
Correlation-screened	21	0.85 ± 0.03	6.4 ± 0.5	16.2
Top-10 permutation	10	0.85 ± 0.03	6.4 ± 0.5	16.2

Table 5: Permutation-importance ranks of the leading descriptors across the seven models (hold-out test set) [VALUES FROM perm_ranks_all_models.csv]

Descriptor	LR	KNN	ANN	Cubist	RF	XGB	HistGB
AD (dose)	27	1	29	1	1	1	1
Time	19	3	1	2	2	2	2
C_i	22	4	14	3	3	4	3
pH	25	2	4	5	7	5	4
BET	24	5	2	7	6	6	6
pH _{pzc}	26	7	19	9	4	7	5
d_{003}	8	6	8	8	12	8	10
Avg. electronegativity	15	13	7	15	9	9	8

Table 6: Machine learning compilations for adsorption prediction [ALL CITATION ROWS REQUIRE VERIFICATION BY THE AUTHORS BEFORE SUBMISSION]

Study	Adsorbent / target	Data	Best model	Repo
[TODO-verify]	biochar/AC, nutrient Q_m	~1,029 records	Cubist	[TO]
[TODO-verify]	[TODO: ML adsorption example 1]	[TODO]	[TODO]	[TO]
[TODO-verify]	[TODO: ML adsorption example 2]	[TODO]	[TODO]	[TO]
This work	LDH, micropollutant RR%	1,342 records / 59 studies	XGB/HistGB	R^2 0.

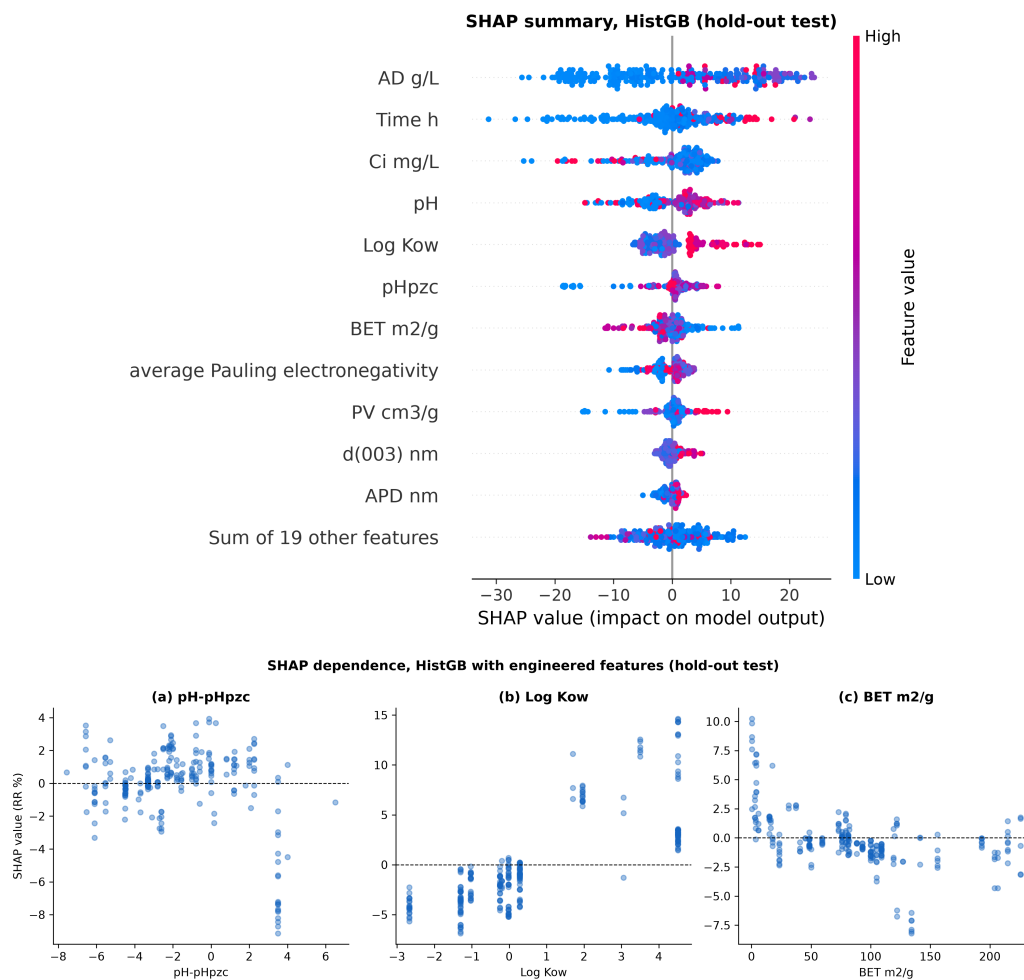


Figure 9: SHAP summary (top) and dependence of mechanism-linked features (bottom)